Intro to Cyber

Program Code: XE101

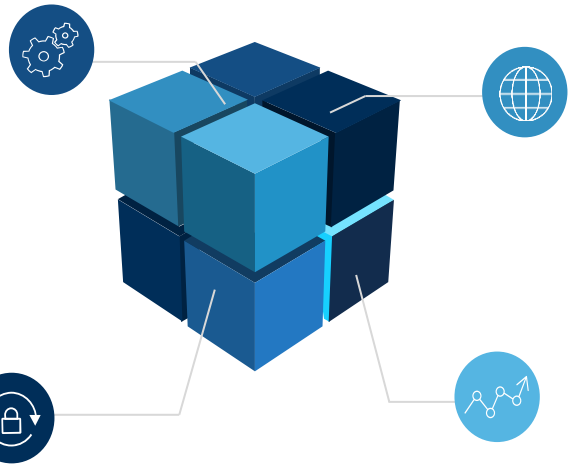
OPERATION ORDER

Operation *Net Crafts* is a two-phased reconnaissance mission. Phase 1, "Network Mapping," commands a detailed survey of the internal network terrain, identifying all devices, their communication protocols, and strategic infrastructure points. Phase 2, "External Intel Gathering," deploys digital surveillance via Shodan and WHOIS, examining the network's public presence and analyzing traffic for operational security. Execute with precision to secure a comprehensive battlefield overview.



1. VISION

Understanding the network, the devices in it, and the role of each of them.



2. MISSION

Map the network and understand the role of each device on your network.

3. STRATEGY

Use an online website such as app.diagrams.net and create a schema of your network.

4. OBJECTIVES

- Find all your IP addresses at home, including MAC addresses.
- Include the DNS and DHCP devices.
- Collect all the details about your ISP and public IP address in Shodan and Whois.

Project Structure

1. Map the Network

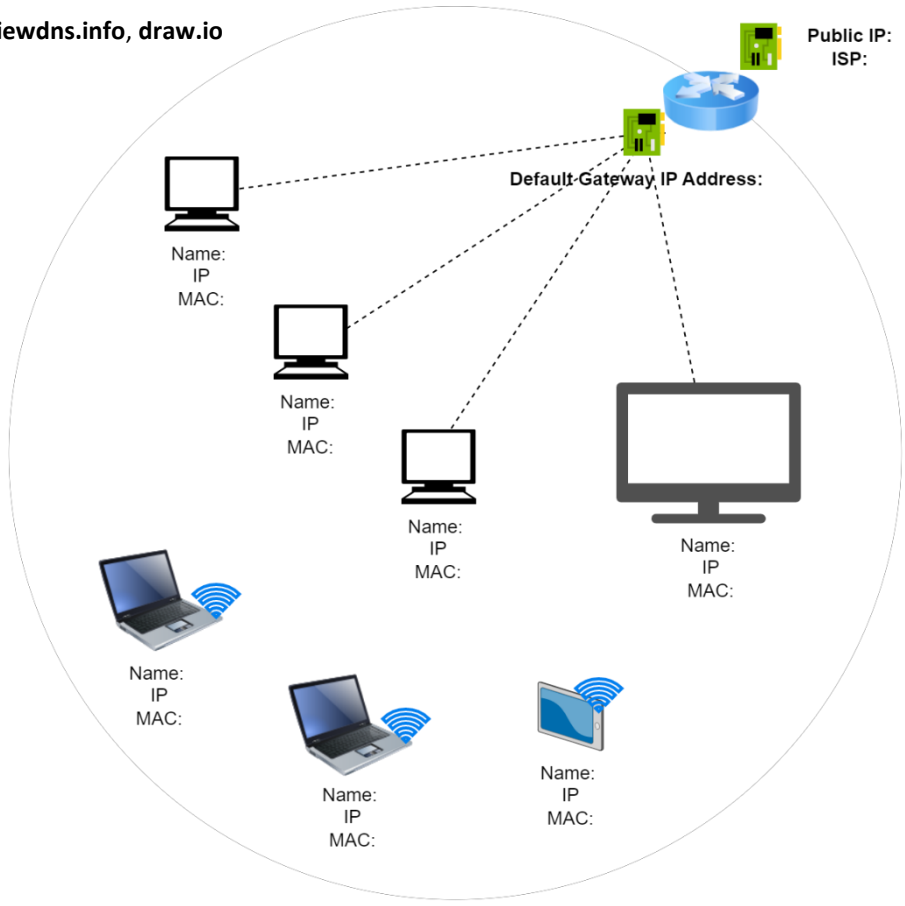
1.1. Display Devices IP Address.
1.2. Display Devices MAC Address and Vendor (display partially).
1.3. Display the Router's Internal and External IP Addresses (display partially).
1.4. Display Device Names.
1.4. Display the DNS and DHCP IP Addresses in your Network.
1.5. Display your Internet Service Provider (ISP).
1.6. Display if the Device is Connected via Ethernet or Wireless.

2. Collecting Information

2.1. Use Shodan to Check Your Public IP Address.
2.2. Use WHOIS to Check Who is Registered on Your Public IP Address.
2.3. Sniff Your Network and Identify Three (3) Used Protocols.
2.3.1. For Each Protocol Explain Its Usage.
2.3.2. For Each Protocol, Find the Used Port Number.

General

Websites available: **shodan.io**, **viewdns.info**, **draw.io**

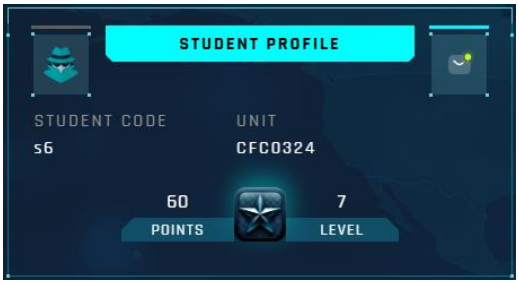


Submitting Your Project

To ensure a successful submission of your project, please adhere to the instructions provided below. Following these guidelines will facilitate a smooth evaluation process and allow you to receive feedback on your work promptly.

1. **What to Submit**
   Submit a PDF document with your network schema and the network details.

2. **File Naming**
   Ensure your project files follow the correct naming convention:
   UNIT.STUDENT.PROGRAM



   For example, if your unit is CFC0324, your student ID is s6, and your program code is xe101, the correct filename should be: CFC0324.s6.xe101.pdf

3. **Submission Process**

   Email Address: Forward your project materials to **projects@thinkcybergroup.com** and to your trainer.

   Email Subject: Format your email subject as follows: **Project: Net Crafts <Student Name:Unit Name>**. Make sure to replace **<Student Name>** with your actual name and **<Unit Name>** with the precise name of the unit as shown in the simulator.

   Response Time: After submitting, you can track the status of your submission (pass or fail) through the ***Progress Tracker*** inside your student account. Allow up to one week for your project to be reviewed.

How to Showcase Your Cybersecurity Projects on LinkedIn

1. **Write a Compelling Post**:
   When you share your project on LinkedIn, make sure to write a post that highlights the importance of the project and your key takeaways from it. Use engaging language and explain:
   - **The challenge** you solved.
   - **Tools** and technologies you used (mention relevant tools like Python, Shell Scripting, Scapy, etc.).
   - **Skills** you developed (problem-solving, incident response, penetration testing, etc.).
   - **Real-world relevance** of your project. Explain how it can be applied to secure networks, detect vulnerabilities, or analyze threats in a professional setting.

Example:

"Excited to share my latest project in Network Security (ZX305) where I built a script to detect network anomalies. This project helped me understand intrusion detection and how to safeguard systems against real-world threats. Using Python and Shell scripting, I developed a tool that can assist security teams in monitoring and reacting to suspicious network activities."

2. **Attach Your PDF**:
   Upload the PDF file of your project as an attachment to your post. This will allow recruiters, potential employers, or other professionals in the cybersecurity field to directly review your work.
   - Ensure your **PDF is professional**, free of errors, and clearly explains each section of the project.
   - **Highlight results**: If you were able to detect or simulate cyber threats, mention how your tool worked and the outcomes of your project.

3. **Record a Video Demo**:
   Videos are powerful on LinkedIn. Record a quick demo of your project where you:
   - Explain the problem.
   - Walk through your code.
   - Show your script in action (for example, showing how your network security tool detects a threat).

Example Intro:

"Here's a quick demo of my project where I developed a Python-based tool to monitor network traffic and detect potential cyber threats. Watch how it scans real-time traffic and flags suspicious activity."

4. **Mention the Certificate**:
   Once you complete the project, your simulator generates an official **ThinkCyber certificate**. Share this alongside your post to demonstrate your verified completion of the project.

Example:

"Proud to have completed my Network Security (ZX305) program with ThinkCyber! This certificate is proof of the skills I've acquired in identifying and mitigating cybersecurity threats."



Make sure to upload the certificate as an image. Use relevant hashtags like **#cybersecurity**, **#networksecurity**, and **#Python** to increase your visibility.

5. **Encourage Networking**:
   Mention that you are open to connecting with cybersecurity professionals and recruiters who might find your skills and projects relevant to their needs.